In [137…
```
pip install bs4
```

```
Requirement already satisfied: bs4 in c:\users\mania\anaconda3\lib\site-packages
(0.0.2)
Requirement already satisfied: beautifulsoup4 in c:\users\mania\anaconda3\lib\sit
e-packages (from bs4) (4.12.3)
Requirement already satisfied: soupsieve>1.2 in c:\users\mania\anaconda3\lib\site
-packages (from beautifulsoup4->bs4) (2.5)
Note: you may need to restart the kernel to use updated packages.
```

In [139…
```
pip install requests
```

```
Requirement already satisfied: requests in c:\users\mania\anaconda3\lib\site-pack
ages (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\mania\anacond
a3\lib\site-packages (from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\mania\anaconda3\lib\site-
packages (from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\mania\anaconda3\lib
\site-packages (from requests) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\mania\anaconda3\lib
\site-packages (from requests) (2024.8.30)
Note: you may need to restart the kernel to use updated packages.
```

In [141…
```
import math
```

In [143…
```
math.factorial(5)
```

Out[143…
```
120
```

In [145…
```
from math import factorial,pow
```

In [147…
```
factorial(5)
```

Out[147…
```
120
```

In [149…
```
pow(2,5)
```

Out[149…
```
32.0
```

In [151…
```
math.pow(2,3)
```

Out[151…
```
8.0
```

In [153…
```
from bs4 import BeautifulSoup
```

In [155…
```
fp = open("sample.html","r")
```

In [159…
```
html = fp.read()
```

In [163…
```
print(type(html))
```

```
<class 'str'>
```

In [ ]:
```
Person p1 = new Person(,)

p1 = Person(,)
```

In [165…
```
soup = BeautifulSoup(html,"html.parser")
```

In [169…
```
soup.find("title").text
```

Out[169…
```
'Online Book Store'
```

In [177…
```
for tag in soup.find_all("h2"):
    print(tag.text)
```

```
Python Programming
Data Science with Python
Machine Learning Basics
```

In [187…
```
for pr in soup.find_all("p", class_ = "price"):
    print(pr.text[3:] + " INR")
```

```
450 INR
650 INR
700 INR
```

In [195…
```
for tag in soup.find_all("div"):
    book = tag.h2.text
    price = tag.find("p", class_ = "price").text[3:]
    print(book + " : " + price)
```

```
Python Programming : 450
Data Science with Python : 650
Machine Learning Basics : 700
```

In [197…
```
import requests
```

In [199…
```
url = "https://quotes.toscrape.com/"
```

In [201…
```
response = requests.get(url)
```

In [203…
```
response.status_code
```

Out[203…
```
200
```

In [209…
```
html_code = response.text
```

In [211…
```
soup = BeautifulSoup(html_code,"html.parser")
```

In [215…
```
soup.find("span", class_ = "text").text
```

Out[215…
```
'“The world as we have created it is a process of our thinking. It cannot be ch
anged without changing our thinking.”'
```

In [219…
```
for tag in soup.find_all("span", class_ = "text"):
    print(tag.text + "\n")
```

“The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking.”

“It is our choices, Harry, that show what we truly are, far more than our abilities.”

“There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle.”

“The person, be it gentleman or lady, who has not pleasure in a good novel, must be intolerably stupid.”

“Imperfection is beauty, madness is genius and it's better to be absolutely ridiculous than absolutely boring.”

“Try not to become a man of success. Rather become a man of value.”

“It is better to be hated for what you are than to be loved for what you are not.”

“I have not failed. I've just found 10,000 ways that won't work.”

“A woman is like a tea bag; you never know how strong it is until it's in hot water.”

“A day without sunshine is like, you know, night.”

In [229…
```
ans = []
for ch in soup.find_all("a", class_ = "tag"):
    if ch.text.startswith("c"):
        ans.append(ch.text)
print(ans)
```

```
['change', 'choices', 'classic']
```

In [237…
```
l1 = []
for ch in soup.find_all("span", class_ = "tag-item"):
    l1.append(ch.text.strip())
```

In [239…
```
l1.sort()
```

In [241…
```
l1
```

Out[241…
```
['books',
 'friends',
 'friendship',
 'humor',
 'inspirational',
 'life',
 'love',
 'reading',
 'simile',
 'truth']
```

In [1]:
```
from bs4 import BeautifulSoup
```

In [3]:
```
fp = open("sample-2.html","r")
```

In [5]:
```
html = fp.read()
```

In [7]:
```
print(html)
```

```
<!DOCTYPE html>
<html>
<head>
    <title>Book Store</title>
</head>
<body>

<h1>Online Book Store</h1>

<div class="book featured" id="b1">
    <h2>Python Basics</h2>
    <p class="price">â‚¹450</p>
    <a href="python.html">View Book</a>
</div>

<div class="book" id="b2">
    <h2>Data Science</h2>
    <p class="price">â‚¹700</p>
    <a href="datascience.html">View Book</a>
</div>

<div class="book" id="b3">
    <h2>Machine Learning</h2>
    <p class="price">â‚¹900</p>
    <a href="ml.html">View Book</a>
</div>

<div class="offer">
    <p>20% Discount Today!</p>
</div>

</body>
</html>
```

In [9]:
```
soup = BeautifulSoup(html, "html.parser")
```

CSS Selector

By tag

In [17]:
```
for tag in soup.select("h2"):
    print(tag.text)
```

```
Python Basics
Data Science
Machine Learning
```

By class

In [24]:
```
for tag in soup.select(".price"):
    print(tag.text[3:])
```

```
450
700
900
```

By id

In [27]:
```
for tag in soup.select("#b1"):
    print(tag.text)
```

```
Python Basics
â,¹450
View Book
```

By tag + class

In [32]:
```
for tag in soup.select("div.book"):
    print(tag.text)
```

```
Python Basics
â,¹450
View Book


Data Science
â,¹700
View Book


Machine Learning
â,¹900
View Book
```

By Multiple Classes

In [45]:
```
for tag in soup.select("a"):
    print(tag["href"])
```

```
python.html
datascience.html
ml.html
```

In [49]:
```
for tag in soup.select("div.book.featured"):
    print(tag.text)
```

```
Python Basics
â,¹450
View Book
```

By Descendent Selector

In [62]:
```
for tag in soup.select("div.book p"):
    print(tag)
```

```
<p class="price">â,¹450</p>
<p class="price">â,¹700</p>
<p class="price">â,¹900</p>
```

By Child Selector

In [67]:
```
for tag in soup.select("div.book > p.price"):
    print(tag.text)
```

```
â,¹450
â,¹700
â,¹900
```

In [ ]:
```
find()
find_all()
```

In [ ]:
```
select()
select_one()
```

In [69]:
```
soup.select_one("h2")
```

Out[69]:
```
<h2>Python Basics</h2>
```

In [71]:
```
from bs4 import BeautifulSoup
```

In [73]:
```
import requests
```

In [75]:
```
url = "https://quotes.toscrape.com/"
```

In [77]:
```
response = requests.get(url)
```

In [79]:
```
html = response.text
```

In [81]:
```
soup = BeautifulSoup(html, "html.parser")
```

In [93]:
```
li = []
for i in soup.select("div.col-md-4.tags-box > span>a"):
    li.append(i.text)
li.sort()
print(li)
```

```
['books', 'friends', 'friendship', 'humor', 'inspirational', 'life', 'love', 'rea
ding', 'simile', 'truth']
```

In [ ]:
```

```